Example by Bijection Principle

- How many bit strings of length *n*, contain exactly *k* 1s?
- **Solution**
  $C(n, k)$.
- How many subsets of a set S with *n* elements, contain exactly *k* elements?
- **Solution**
  We have already seen that there is a 1:1 correspondence between the subsets of *k* elements of a set *S* with cardinality *n*, with the bit strings of length *n* that contain exactly *k* 1s. Thus, the answer is the same as the first question $C(n, k)$.